A-A-340A
April 29, 1985
SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

INSECTICIDE; MIXTURE OF D-TRANS ALLETHRIN AND RESMETHRIN

The General Services Administration has authorized the use of this commercial item description.

This commercial item description covers an insecticide space spray for use in enclosed areas against mosquitoes, flies, gnats, and other small flying insects; it may also be sprayed directly on insects.

Salient characteristics: The insecticide shall be a water-based product packaged in a push-button, pressurized spray can and shall conform to the following requirement when tested as specified.

Resmethrin[5-(phenylmethyl)-3-furanyl]methyl-2,2-dimethyl-3-(2-methyl-1propenyl) cyclopropanecarboxylate*	Minimum 0.200% (w/w)
plus	
d-Trans Allethrin (allyl homolog of Cinerin I)	Minimum 0.150% (w/w)

*cis/trans isomer ratio: max 30% cis and min 70% trans

Other additional active ingredients may include related compounds of the above toxicants, aromatic petroleum hydrocarbons, and aliphatic petroleum distillates.

Spraying properties. The pressurized container shall deliver a minimum of 98 percent by weight of the contents when tested as specified. The valve shall spray easily with moderate finger pressure and close immediately upon release of pressure. The spray shall be uniform and without sputtering.

Net contents 300 grams (10.6 ounces) minimum

Test methods:

RESMETHRIN/d-TRANS ALLETHRIN CONTENT

The resmethrin/d-trans allethrin content shall be reported as the average of determinations made on three randomly selected containers.

FSC 6840

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1. Apparatus - Gas chromatograph equipped with flame ionization detector and suitable integrator. Conditions may be modified, if necessary, to obtain proper separations, reproducibility, and sensitivity. Operating conditions shall be reported as a part of the test report.

Recommended Chromatographic Conditions

CHARACTERISTIC	CONDITION
Column Material	Glass
Column Dimensions	2 m X 4 mm. I.D.
Column Support	80/100 mesh Chromosorb W(HP)
Liquid Phase	5% SE-30
Injector Temperature	250°C
Column Temperature	220°C for 10 min., and then programmed to 260°C at 10°C/min.
Detector Temperature	250°C
Carrier Gas	Helium
Carrier Gas Flow Rate	30 mL per min.

2. Internal Standard Solution - Dissolve 1 gram docosane in choroform, transfer to a 200-mL volumetric flask and dilute to the mark with chloroform.

3. Standard Solution - Transfer about 30 mg of analytical standard resmethrin and 25 mg d-trans allethrin accurately weighed to a 10-mL volumetric flask and dissolve in about 2 mL chloroform. Add 5.0 mL of the internal standard solution, dilute to the volume with chloroform and mix.

3.1 Analytical standard resmethrin may be obtained from S.B. Penick & Co., Commercial Development - Pesticides, 215 Watchung Avenue, Orange, NJ 07050. Analytical standard d-Trans Allethrin may be obtained from McLaughlin, Gormley, King Co., 8810 Tenth Avenue North, Minneapolis, MN 55427.

4. Sample Preparation - Weigh aerosol can and place in a dry ice-acetone bath or freezer until contents solidify. Remove container from bath, punch two small holes in the bottom, and allow propellant to evaporate in a fume hood. Do not apply extra heat. Quantitatively transfer the liquid to a 500-mL volumetric flask, rinse the can with water and dilute to volume. Pipette a 25 mL aliquot into a 100-mL round bottom flask and remove the water using a rotary evaporator at 60°C under vacuum.

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If the emulsion is too thick, then approximately 25 grams may be weighed into the flask instead of using a pipette. To the oily residue in the flask add 5.0 mL of internal standard solution, 5.0 mL of chloroform and mix.

5. Analysis Procedure - Condition the column before use by injection of the standard solution until the peak height ratios of the first peak, d-trans allethrin, second peak, docosane, and the third peak, resmethrin, are constant.

Inject 1-3 uL of the standard solution and calculate the response factors as follows:

$$F_1 = \frac{AB}{CDE} \qquad \text{and} \qquad F_2 = \frac{AG}{HIE}$$

Where: F_1 & F_2 = Response factor of d-trans allethrin & resmethrin, respectively

A = Weight of internal standard (docosane) in mg

B & G = Areas of d-trans allethrin & resmethrin reference standard peaks, respectively

C & H = purity of d-trans allethrin & resmethrin standard, respectively

D & I = Weight of d-trans allethrin & resmethrin, respectively, in mg

E = Area of internal standard (docosane) peak

Inject 1-3 uL of the sample preparation and calculate the % d-trans allethrin and resmethrin as follows:

$$\% \text{ allethrin} = \frac{AB(100)}{CW\ F_1} \qquad \% \text{ resmethrin} = \frac{AE(100)}{CW\ F_2}$$

Where: A = Weight of internal standard (docosane) in mg
B & E = Areas of d-trans allethrin & resmethrin peaks, respectively
C = Area of internal standard peak (docosane)
W = Weight of sample aliquot in mg
F_1 & F_2 = Response factor for d-trans allethrin & resmethrin, respectively

$$\text{Weight of sample aliquot(W) in mg} = \frac{(B - A)\ 25}{500} \times 1000$$

Where: T = Total weight of aerosol can before opening
S = Weight of empty can

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SPRAYING PROPERTIES

Samples for the spraying properties test shall be selected in accordance with MIL-STD-105, inspection level S-2, with an AQL of 4.0 percent defective.

Remove the protective cap which covers the dispensing valve. Weigh the dispenser and record the gross weight to the nearest tenth of a gram. Hold the dispenser upright and exhaust the container by holding the valve wide open. During this action, the temperature of the can will drop; when this occurs, allow the can to return to room temperature before proceeding to exhaust any additional insecticide. When no further insecticide is expelled, weigh the can and record the weight, again to the nearest tenth of a gram. Open the can, drain off any remaining insecticide, and allow the can to dry. Reweigh the empty container and calculate percent delivered as follows:

$$\frac{T - X}{T - S} \times 100 = \% \text{ by weight of contents delivered.}$$

Where: T = weight in grams of full can.

X = weight in grams of can after spraying.

S = weight in grams of empty can after thoroughly dry.

Check the percent for compliance with the above requirements.

Net contents

Calculate the net contents of each container used in the spraying properties test to determine compliance with the minimum net weight requirement. An AQL of 4.0 percent defective shall be used for this test.

Regulatory requirements. The product shall be labeled and registered with the Environmental Protection Agency in accordance with 40 CFR 162 - Regulations for the Enforcement of the Federal Insecticide, Fungicide, and Rodenticide Act. The product shall not contain chlorofluorocarbons in accordance with 40 CFR 762.

Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be as specified in the contract or order.

Notes. The Code of Federal Regulations (CFR) is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402

Reviewing activity:
DLA-GS

Preparing activity:
GSA-FSS

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